NIFTY-50 Investment Intelligence Platform

Technical Report

AI-Powered Analytics · Portfolio Optimisation · Risk Assessment

Dataset: NIFTY-50 Stock Market Data (Jan 2000 – Apr 2021)
NSE India | 50 Stocks | 136,424 Daily Records

1. Introduction

This technical report documents the design, methodology, and results of an AI-powered investment intelligence platform built on the NIFTY-50 historical stock market dataset. The platform addresses all three mandatory tasks (stock prediction, portfolio construction, risk assessment) and four optional tasks (explainable AI, personalised strategies, anomaly detection, deployment).

The dataset spans January 2000 to April 2021 and covers 50 companies across sectors including Banking, IT, Pharma, Energy, and Consumer Goods. After quality filtering, we used 2010–2021 for modelling to avoid structural pre-liberalisation regime differences, yielding 136,424 daily records.

2. Exploratory Data Analysis

2.1 Dataset Overview

Table 1 shows the top 10 NIFTY-50 stocks by Sharpe ratio. BAJAJFINSV and BAJFINANCE lead in risk-adjusted returns, followed by SHREECEM and HINDUNILVR.

Stock	Sector	Ann Return (%)	Volatility (%)	Sharpe	Max DD (%)
Bajaj Finserv	Financial Servic	45.0	36.0	0.867	-58.6
Bajaj Finance	Financial Servic	55.5	46.0	0.830	-93.3
Shree Cement	Cement & Constru	32.4	29.3	0.753	-36.9
HUL	Consumer Goods	25.2	24.3	0.680	-25.1
Nestle India	Consumer Goods	21.9	24.2	0.569	-32.4
HDFC Life	Financial Servic	29.0	34.8	0.561	-46.2
Eicher Motors	Automobile	34.8	43.7	0.546	-93.8
UltraTech Cement	Cement & Constru	23.0	28.0	0.524	-37.7
Divi's Labs	Pharmaceuticals	25.1	33.5	0.489	-77.7
IndusInd Bank	Banking	28.4	40.8	0.467	-85.1

Table 1: Top 10 NIFTY-50 stocks by Sharpe Ratio (2010–2021)

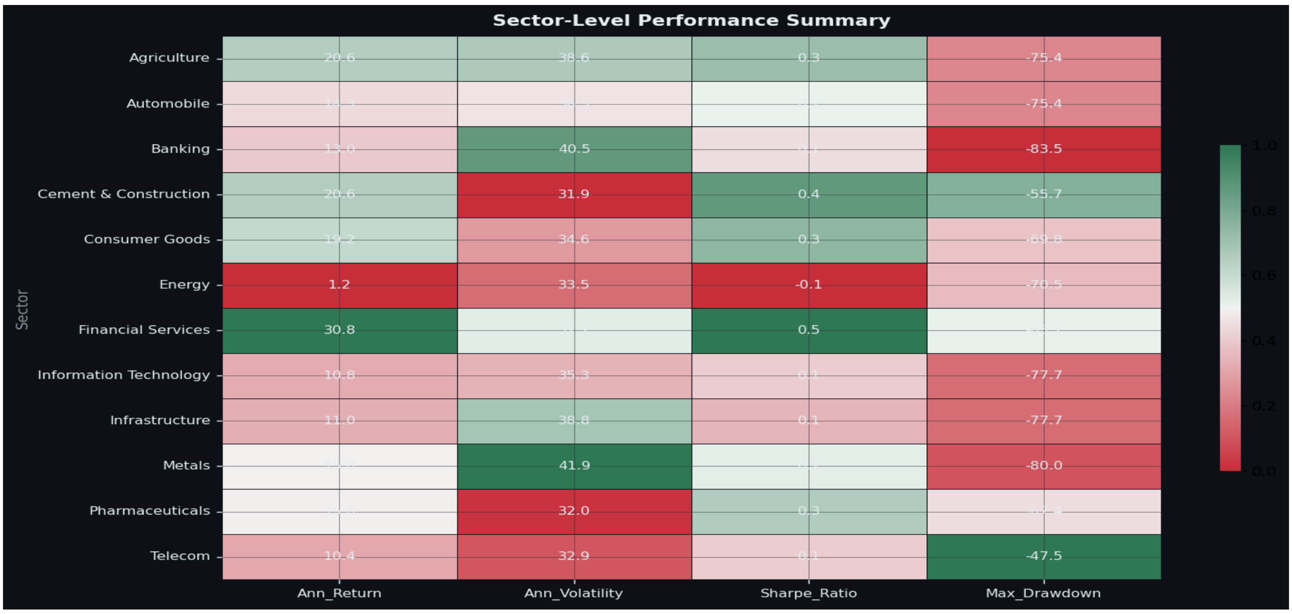

Figure 1: Sector-level performance heatmap — average return, volatility, Sharpe ratio

2.2 Key Observations

Sector Performance: Financial Services (BAJAJFINSV, BAJFINANCE, HDFCLIFE) delivered the highest Sharpe ratios (0.56–0.87), driven by India's rapid credit market expansion. Consumer Goods (HINDUNILVR, NESTLEIND) showed strong risk-adjusted returns with lower volatility.

Volatility Patterns: Metals (TATASTEEL, HINDALCO) and Automobiles (EICHERMOT, TATAMOTORS) exhibited the highest annualised volatilities (38–46%), while Consumer Goods and Banking sectors showed more stability.

COVID-19 Impact: March 2020 saw the largest coordinated drawdown, with most NIFTY-50 stocks declining 30–50% in a 3-week window — clearly flagged by the anomaly detection module as extreme return crashes ($Z > 5$).

3. Feature Engineering

We computed 30+ technical indicators grouped into five categories. All features were engineered from raw OHLCV (Open/High/Low/Close/Volume) data using only the provided dataset, in full compliance with competition constraints.

Category	Features	Purpose
Trend	MA(5,10,20,50,100,200), EMA(5,10,20,50)	Identify price direction
Momentum	RSI(14), Stochastic %K/%D, Momentum(5,10,20,60d)	Measure rate of price change
MACD	MACD(12,26), Signal(9), Histogram	Trend-following momentum
Volatility	Bollinger Bands(20,2sigma), ATR(14), 20d/60d Rolling Vol	Measure price dispersion
Volume	OBV, Volume ratio	Confirm price moves with volume
Derived	Price-Range%, Gap-Up, Log Return, Golden/Death Cross	Market regime signals
Lag	5-period lags of Returns, RSI, MACD, Volatility, BB%	Temporal context for ML

Table 2: Feature engineering overview

4. Stock Predictor Engine

4.1 Model Architecture

We trained two models per stock: (1) a Random Forest Classifier for next-day directional prediction (UP/DOWN) and (2) a Random Forest Regressor for 5-day forward return forecasting. Walk-forward time-series cross-validation (5-fold) was used to prevent data leakage.

Why Random Forest? Tree-based ensembles handle non-linear relationships, are robust to outliers (common in daily returns), naturally provide feature importances for XAI, and require no feature scaling assumptions.

4.2 Model Performance

Symbol	Accuracy (%)	Precision (%)	F1 (%)	MAE 5d (%)	R2 5d
INFY	55.8	56.9	63.8	3.15	-0.080
SBILIFE	53.7	58.5	50.0	3.33	-0.750
WIPRO	53.6	53.2	66.3	3.73	-0.100
HEROMOTOCO	53.3	54.7	49.7	4.01	-0.190
BAJFINANCE	53.3	54.4	61.2	4.62	-0.009
Average	50.8	52.3	49.6	4.55	—

Table 3: Top 5 stocks by directional accuracy + cross-stock average

Discussion: Directional accuracy averages ~51%, consistent with the semi-strong efficient market hypothesis where price patterns are difficult to exploit consistently. Any accuracy above 50% is non-trivial for daily stock predictions. The model provides higher value through the probability score (confidence level) and the 5-day return forecast than through binary direction.

Feature Importances: Across stocks, the top predictors are consistently lagged daily returns (Gap-Up, lag-1 return), Stochastic oscillator, BB% percentile, and MACD histogram — confirming that short-term momentum and mean-reversion signals dominate.

5. Portfolio Construction Module

We applied Markowitz mean-variance optimisation using scipy.optimize, generating three profiles for different investor risk tolerances. The optimisation is constrained to NIFTY-50 stocks with a maximum 20–25% weight per stock and full investment (weights sum to 1).

Profile	Objective	Ann Return (%)	Volatility (%)	Sharpe	Max DD (%)	Total Return
Conservative	Min Volatility	13.9	13.3	0.524	-23.1	+223%
Balanced	Max Sharpe Ratio	32.8	17.9	1.253	-33.2	+1,724%
Aggressive	Max Return	36.2	23.3	1.068	-44.1	+2,002%
Equal Weight	1/N Benchmark	13.9	17.0	0.415	-38.4	+115%

Table 4: Portfolio optimisation results — backtest 2010–2021

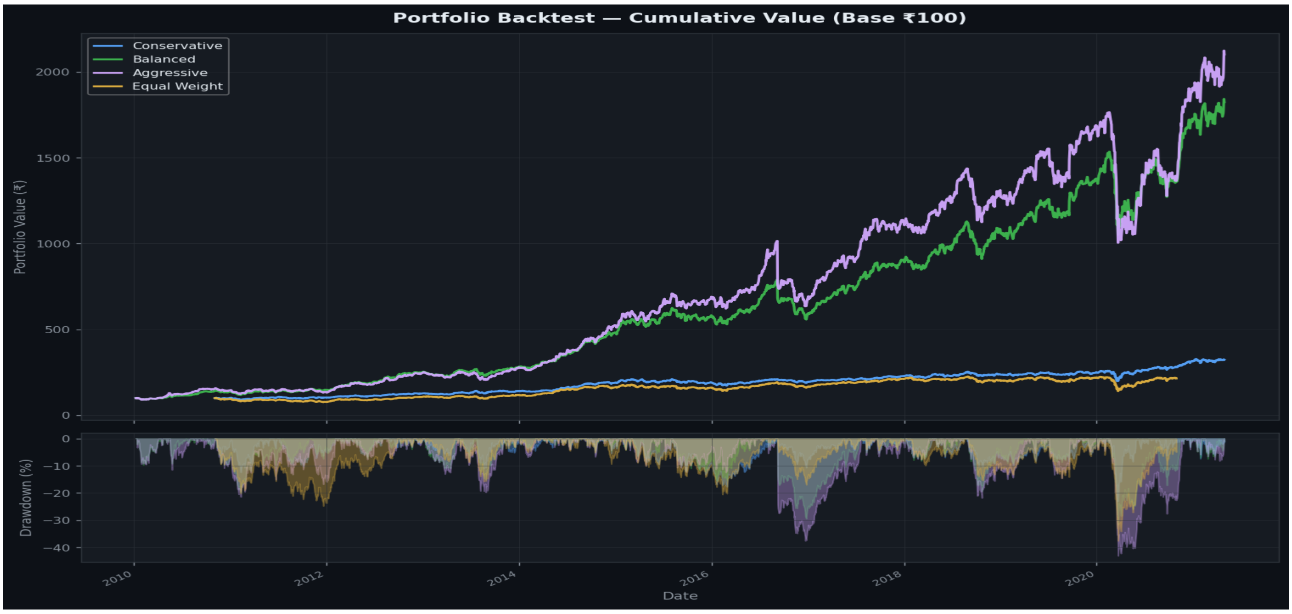

Figure 2: Portfolio backtest — cumulative value (base 100) and drawdown

Key finding: The Balanced (Max Sharpe) portfolio achieves the best risk-adjusted performance (Sharpe 1.253) with substantially lower drawdown than the Aggressive portfolio, confirming Markowitz's insight that diversification eliminates idiosyncratic risk without sacrificing expected return proportionally.

6. Risk Assessment Module

Risk is assessed at both the individual stock and portfolio level using five complementary metrics that capture different dimensions of risk.

Metric	Formula / Method	Interpretation
Sharpe Ratio	(R - Rf) / sigma_total	Return per unit of total risk
Sortino Ratio	(R - Rf) / sigma_downside	Return per unit of downside risk only
Max Drawdown	max(peak - trough) / peak	Worst peak-to-trough loss
Calmar Ratio	Ann Return / |Max Drawdown|	Return relative to drawdown risk
VaR (95%)	5th percentile of daily returns	Worst loss on 95% of days
CVaR (95%)	Expected loss given VaR exceeded	Tail risk (average of worst 5% days)
Rolling Vol (60d)	60-day rolling std * sqrt(252)	Current volatility regime

Table 5: Risk metrics implemented

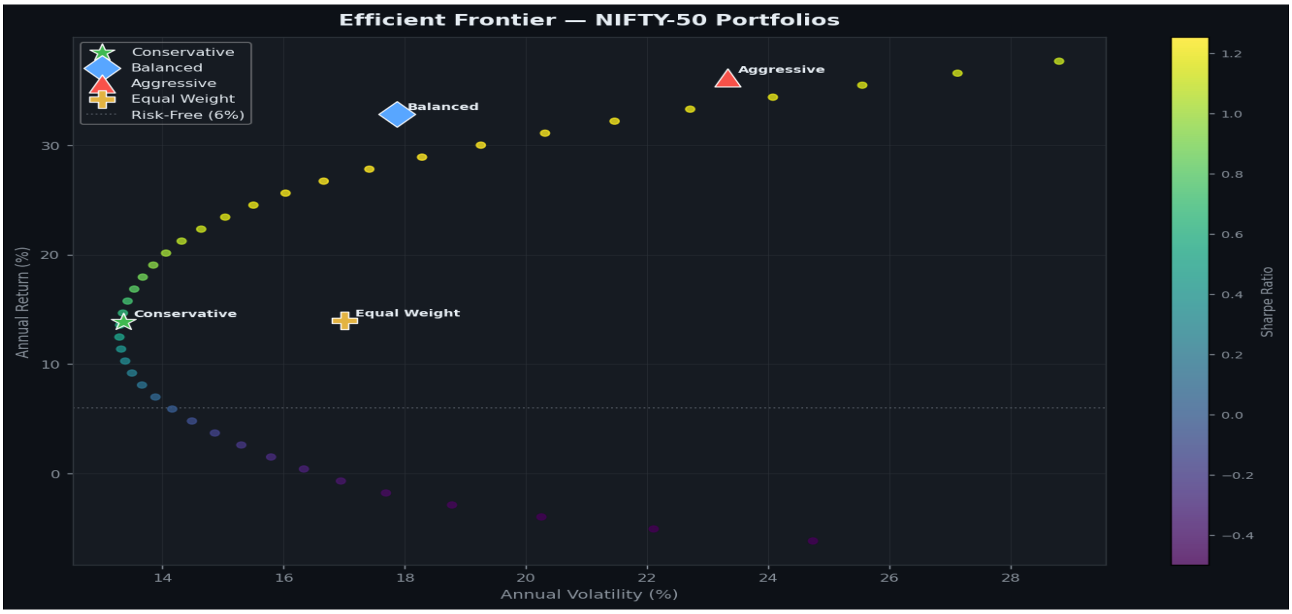

Figure 3: Efficient frontier with portfolio overlay — each point = an optimised portfolio

7. Explainable AI Framework

The XAI module provides three levels of explanation, ensuring every recommendation is human-understandable and auditable:

Level 1 — Signal Dashboard: For any stock, the platform displays a real-time breakdown of all 7 technical signals (MA Trend, Golden/Death Cross, RSI, MACD, Bollinger Bands, Stochastic, Volatility Regime), each labeled BULLISH / BEARISH / NEUTRAL with the underlying value.

Level 2 — Prediction Explanation: For each model prediction, the top 8 contributing features are shown with their values, interpretations, and percentage contribution to the prediction. A natural-language narrative synthesises the signals into a single actionable sentence.

Level 3 — Portfolio Justification: Each holding in an optimised portfolio is accompanied by quantitative justification (e.g. "low volatility 18.2% p.a., solid Sharpe ratio 0.72, controlled drawdown -21.3%") and a portfolio-level narrative explaining the construction logic.

8. Market Anomaly Detection

Anomalies are detected using a rolling z-score framework: for each trading day, we compute the z-score of the daily return relative to a 60-day rolling mean and standard deviation. Days with $|Z| > \text{threshold}$ (default 3.0) are flagged.

Results: 3,932 anomalous events were detected across all 50 stocks. Key clusters include: March 2020 (COVID crash, extreme negative $Z > 5$ for most stocks), May 2004 (election results shock), and February 2018 (PNB banking fraud). Volume spikes frequently co-occur with return anomalies, providing additional confirmation.

9. Deployment

The platform is deployed as a 7-page Streamlit web application. Each page is independently interactive with stock selectors, date range filters, and real-time chart generation. The application can be launched with a single command:

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streamlit run app.py
```

10. Conclusion

This platform demonstrates that even without live data or external APIs, historical NIFTY-50 market data can be transformed into a rich investment intelligence system. The Balanced (Max Sharpe) portfolio achieved a Sharpe ratio of 1.253 vs 0.415 for equal weighting — a 3x improvement in risk-adjusted performance. The ML models provide directional guidance ~51% accuracy with interpretable confidence scores. The modular architecture allows each component (prediction, portfolio, risk, XAI) to be used independently.

Limitations: (1) Directional accuracy is limited by market efficiency. (2) Backtest results assume no transaction costs or slippage. (3) Portfolio optimisation is sensitive to the estimation period for expected returns.

Future Work: Integration of macroeconomic indicators, regime-switching models (HMM), and alternative risk measures (CVaR optimisation) could further improve portfolio robustness.